Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Alexander, J. Iwan D. Ph.D.		
Project Title:	Vibrations and g-Jitter: Transport Disturbances Due to Residual Acceleration During Low Gravity Directional Solidification Experiments		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):	Kassemi, Mohammad (Case Western Reserve University) Kizito, John (National Center for Microgravity Research on Fluids and Combustion)		
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	The experimental objectives are to characterize the response of heat and species transport in a directionally solidifying model transparent alloy system subject to well-characterized imposed vibrations and ultimately to spacecraft residual acceleration. The temperature and composition fields will be measured using two-color holographic interferometry to characterize the composition and temperature fields simultaneously. The convective flows will be visualized using a shadowgraph technique and selected experiments will involve Particle Image Velocimetry. These experiments will be accompanied by numerical modeling that will be used extend the ground-based experimental results and to assist in the definition of a set of benchmark flight experiments. The experiments will:		

Task Description:	· Quantify the effects of residual acceleration on transport in compositionally inhomogeneous liquids and on transport during directional solidification. · Investigate vibrational convective transport under 1 g conditions in transparent model systems (succinonitrile-water, SCN-W) using High and low frequency translational vibrations and Rotational vibrations. · Investigate single and two-liquid phase melt responses. · Compare thermosolutal and dominantly thermal convective responses to vibration (facilitated by the weak dependence of SCN-W density on composition). · Investigate the possibility of using vibration for transport control (e.g., for melt homogenization or rehomogenization). The objective is to characterize the response of heat and species during directional solidification of a binary alloy system subject to translational and rotational vibration. This will be accomplished through a combination of ground-based experiments and numerical simulation. The work will lead to the definition of benchmark flight experiments that will unambiguously characterize the response of solidification systems to spacecraft residual acceleration in low-earth orbit. Since the ground-based work involves vibrations that either correspond to modulations of 1 g or are lateral vibrations in the presence of steady 1 g acceleration, the emphasis of the ground-based work will be on the generation of mean flows at higher frequencies. The design of experiment facilities and planning of specific experiments for the International Space Station and other orbital platforms requires that the experiment's sensitivity to microgravity residual acceleration be assessed. Residual acceleration arises due to transient and vibrational disturbances (g-jitter), atmospheric drag on the spacecraft, and the gravity gradient. The interaction of residual acceleration with density gradients in fluids will generally produce buoyancy driven flows within those fluids. Experiments involving fluids with density gradients are conducted in low gravity so that buoyant convective motion that is unavoidable under terrestrial conditions is reduced or eliminated. To make best use of the available experiment time and to eliminate any ambiguities in the interpretation of experiment results it is therefore essential to assess the sensitivity of a given experiment to time-dependent and steady acceleration. To date, nearly all the work that addresses the impact of residual acceleration on low-gravity experiments relies on the results of numerical modeling and scaling analyses. Experimental quantification or verification is all but lacking. On the other hand, questions regarding the possible influence of residual acceleration effects on actual flight experiments still persist. The recent MEPHISTO flight experiments (conducted on USMP-3, in February-March 1996) were the first experiments to be conducted to characterize the effects of impulse disturbances (caused by thruster firings) on transport during directional solidification. The experiment results clearly demonstrated the ability of numerical models to predict the convective transport response to impulse disturbances. However, a quantitative characterization of experiment response to other forms of g-jitter acceleration is still lacking, most notably in the area of vibration and rotational acceleration. (Note that for rotational accelerations that involve "pendulum" or rocking type motions, fluid motion can occur even in the absence of density gradients). Without quantitative experimental verification, model predictions of experiment response to g-jitter are of limited use and could lead to unnecessary design restrictions or unsuitable experiment operating conditions. Thus, to further improve our ability to predict the impact of vibration and rotational g-jitter acceleration, quantitative experimental data is needed. This will establish a set of benchmark results that can be used to unambiguously define the response of a well-characterized solidification experiment to residual acceleration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this reporting period.
Bibliography Type:	Description: (Last Updated: 04/21/2017)